

Exercise-2

Problem statement-Implementing the Factory Method Pattern

Source code:

```
using System;

public interface IDocument
{
    void Open();
}

public class WordDocument : IDocument
{
    public void Open()
    {
        Console.WriteLine("Opening Word Document...");
    }
}

public class PdfDocument : IDocument
{
    public void Open()
    {
        Console.WriteLine("Opening PDF Document...");
    }
}
```

```
public class ExcelDocument : IDocument
{
    public void Open()
    {
        Console.WriteLine("Opening Excel Document...");
    }
}

public abstract class DocumentFactory
{
    public abstract IDocument CreateDocument();
}

public class WordDocumentFactory : DocumentFactory
{
    public override IDocument CreateDocument()
    {
        return new WordDocument();
    }
}

public class PdfDocumentFactory : DocumentFactory
{
    public override IDocument CreateDocument()
    {
        return new PdfDocument();
    }
}
```

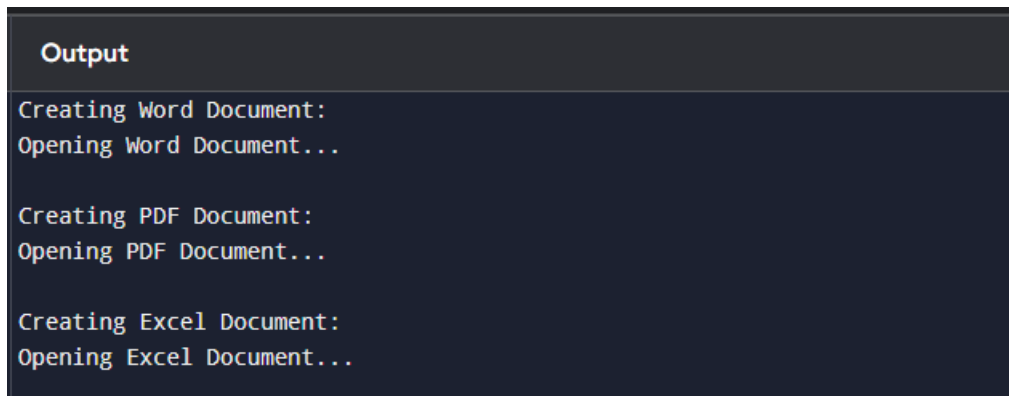
```

    }
}
public class ExcelDocumentFactory : DocumentFactory
{
    public override IDocument CreateDocument()
    {
        return new ExcelDocument();
    }
}
class Program
{
    static void Main(string[] args)
    {
        DocumentFactory factory;
        Console.WriteLine("Creating Word Document:");
        factory = new WordDocumentFactory();
        IDocument word = factory.CreateDocument();
        word.Open();
        Console.WriteLine();
        Console.WriteLine("Creating PDF Document:");
        factory = new PdfDocumentFactory();
        IDocument pdf = factory.CreateDocument();
        pdf.Open();
    }
}

```

```
        Console.WriteLine();  
        Console.WriteLine("Creating Excel Document:");  
        factory = new ExcelDocumentFactory();  
        IDocument excel = factory.CreateDocument();  
        excel.Open();  
    }  
}
```

Output:



```
Output  
Creating Word Document:  
Opening Word Document...  
  
Creating PDF Document:  
Opening PDF Document...  
  
Creating Excel Document:  
Opening Excel Document...
```